

Win-Rate Confidence and Sample Size, Worked Out by Hand

Why “model A wins 6 of 10” proves nothing — and exactly how many comparisons you need.

What this document does. It takes one pairwise A/B result — model A wins 62 of 100 head-to-head judgements — and computes both the normal-approximation and the Wilson 95% confidence interval by hand. Then it inverts the question: for a target margin of error, how many comparisons N do you need? It derives the $N \approx (1.96/E)^2 p(1 - p)$ rule and tabulates it. Every number is reproduced by the program in Appendix A.

Where this sits. This is **Topic 2 of Module 3.5** (human evaluation protocols, pairwise win-rate). Companions: Module 3.5 Topic 1 (Cohen’s κ) and Module 3.6 (LLM-judge validation, which consumes these intervals).

Concepts covered, in order: win-rate as a proportion · standard error · the normal-approx interval · the Wilson interval · inverting for sample size · the worst-case $p = 0.5$ · why small eval sets decide nothing.

Internal learning material. Numbers are reproducible from Appendix A. v1.0

1 The setup: 62 wins out of 100

A human (or judge) compares model A against model B on N prompts and picks a winner each time. Model A wins $k = 62$, so the observed win-rate is $\hat{p} = 0.62$. Is A really better, or is this within coin-flip noise of 0.50?

Quantity	Symbol	Value here
Comparisons	N	100
A wins	k	62
Observed win-rate	$\hat{p} = k/N$	0.62
Confidence	$1 - \alpha$	95% ($z = 1.96$)
Target margins (later)	E	0.10, 0.05, 0.03, 0.01

Notation. A win-rate is a proportion estimated from N Bernoulli trials, so its sampling variance is $\hat{p}(1 - \hat{p})/N$ and its standard error is the square root of that.

Intuition

Every win-rate is an estimate with a cloud of uncertainty around it. The width of that cloud shrinks like $1/\sqrt{N}$ — slowly. Quadrupling your eval set only halves the error bar. So the question is never just “who won” but “is the gap wider than the error bar?”

2 Step 1 — the standard error and the normal interval

$$\text{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{N}} = \sqrt{\frac{0.62 \cdot 0.38}{100}} = \sqrt{0.002356} = 0.04854.$$

The normal-approximation 95% interval is $\hat{p} \pm z \text{SE} = 0.62 \pm 1.96(0.04854) = 0.62 \pm 0.0951$:

$$\text{normal 95\% CI} = [0.5249, 0.7151]. \quad (1)$$

The interval sits *above* 0.50 — so at $N = 100$, A's advantage is (just) statistically real. Had A won only 56, the interval would be $0.56 \pm 0.097 = [0.463, 0.657]$, straddling 0.50: not significant.

Mini-example — this operation in isolation

The same data at $N = 10$ instead of 100: $\hat{p} = 0.6$, $\text{SE} = \sqrt{0.6 \cdot 0.4/10} = 0.155$, $\text{CI} = 0.6 \pm 0.304 = [0.296, 0.904]$. ✓ That interval is enormous and spans 0.5 easily. “A wins 6 of 10” is consistent with A being *worse* than B. Ten comparisons decide nothing.

3 Step 2 — the Wilson interval (better near 0 and 1)

The normal approximation misbehaves for small N or extreme $\hat{p}$ (it can even poke past 0 or 1). The **Wilson score interval** fixes this by solving for the p values consistent with $\hat{p}$:

$$\frac{\hat{p} + \frac{z^2}{2N} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}}}{1 + \frac{z^2}{N}}. \quad (2)$$

For $k = 62$, $N = 100$, $z = 1.96$ this gives $[0.5221, 0.7090]$ — centred slightly below $\hat{p}$ and a touch narrower at the top than the normal interval. Both agree here that $A > B$; the Wilson form is the one to trust when N is small or $\hat{p}$ is near an edge.

normal: [0.525, 0.715]	Wilson: [0.522, 0.709]	both exclude 0.5.
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(3)

4 Step 3 — inverting for sample size

Now plan ahead. Demand that the 95% margin of error be at most E : $z\sqrt{p(1-p)/N} \leq E$. Solve for N :

$$N \geq \left(\frac{z}{E}\right)^2 p(1-p). \quad (4)$$

The variance $p(1-p)$ is largest at $p = 0.5$ (value 0.25), so the safe, worst-case planning number drops the unknown p and uses 0.25: $N \geq (1.96/E)^2 \cdot 0.25 = (0.98/E)^2$.

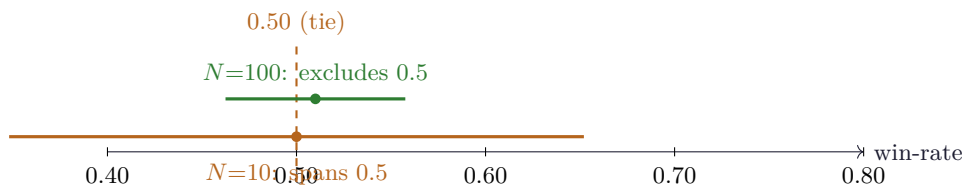
margin E	required N (at $p=0.5$)	reading
0.10	97	coarse: ± 10 pts
0.05	385	the classic “ ~ 400 ” rule
0.03	1,068	polling-grade ± 3 pts
0.01	9,604	± 1 pt costs ~ 10 k judgements

So a $\pm 5\%$ win-rate needs ≈ 385 comparisons; tightening to $\pm 1\%$ costs $25\times$ more. ✓ (Check $E = 0.05$: $(1.96/0.05)^2 \cdot 0.25 = 39.2^2 \cdot 0.25 = 1536.64 \cdot 0.25 = 384.16 \rightarrow 385$.)

Watch out

$N \approx 385$ buys a $\pm 5\%$ error bar, which is not the same as *detecting a small difference*. To call a true 52% win-rate significantly above 50% you need far more — the effect (0.02) is smaller than the error bar at $N = 385$. Distinguishing close models is a *power* question and demands thousands of comparisons.

5 The interval, drawn



Same point estimate (≈ 0.6), wildly different conclusions: the wide $N=10$ bar crosses the tie line; the tight $N=100$ bar clears it.

6 Cost cheat-sheet

Win-rate	$\hat{p} = k/N$
Standard error	$SE = \sqrt{\hat{p}(1 - \hat{p})/N}$
Normal 95% CI	$\hat{p} \pm 1.96 SE$
Wilson 95% CI	Eq. (2) (use for small N /extreme $\hat{p}$)
Sample size	$N \geq (z/E)^2 p(1 - p)$
Worst case ($p=0.5$)	$N \geq (0.98/E)^2$
$\pm 5\%$ at 95%	$N \approx 385$

7 An honest word about these numbers

The 62-of-100 result is illustrative. What is invariant: a win-rate is a proportion, its error falls only as $1/\sqrt{N}$, and the worst-case sample size for margin E is $(0.98/E)^2$. Report the interval next to every win-rate; an A/B headline without N and a CI is a coin flip dressed as a result.

And remember that a tight *error bar* is not the same as the *power* to separate two close models — that is a larger bill again.

Appendix A — Reference implementation

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1  # win_rate_sample_size.py -- reproduces every number in the document.
2  import math
3  z = 1.96
4
5  def normal_ci(k, N):
6      p = k / N; se = math.sqrt(p*(1-p)/N)
7      return p, se, p - z*se, p + z*se
8
9  def wilson_ci(k, N):
10     p = k / N; den = 1 + z*z/N
11     centre = (p + z*z/(2*N)) / den
12     half = (z*math.sqrt(p*(1-p)/N + z*z/(4*N*N))) / den
13     return centre - half, centre + half
14
15 p, se, lo, hi = normal_ci(62, 100)
16 print(f"p={p:.2f} SE={se:.5f} normal CI=[{lo:.4f}, {hi:.4f}]")
17 print(f"wilson CI=[{wilson_ci(62,100)[0]:.4f}, {wilson_ci(62,100)[1]:.4f}]")
18 p2, se2, lo2, hi2 = normal_ci(6, 10)
19 print(f"N=10: p={p2:.1f} CI=[{lo2:.4f}, {hi2:.4f}] (spans 0.5)")
20
21 for E in (0.10, 0.05, 0.03, 0.01):
22     N = (z/E)**2 * 0.25
23     print(f"E={E:.2f} -> N={math.ceil(N)}")
24
25 # Expected output:
26 # p=0.62 SE=0.04854 normal CI=[0.5249, 0.7151]
27 # wilson CI=[0.5221, 0.7090]
28 # N=10: p=0.6 CI=[0.2964, 0.9036] (spans 0.5)
29 # E=0.10 -> N=97
30 # E=0.05 -> N=385
31 # E=0.03 -> N=1068
32 # E=0.01 -> N=9604

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